The quadratic memory core in the double-scaled limit
A strong filter limit from sixth Racah moments

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Abstract

We complete the ordinary Gaussian second-chaos limit of the quartic memory core at half filling. A uniform sixth moment of the positive rank-product distribution controls a growing output window. Together with the preceding coherent norm calculation, it proves the missing A-filter replacement in L^2. The core then has an explicit expansion in four directions and a strictly positive residual after projecting onto the three previously retained directions.

1 Setting and completed core

Use standard CAR, the full overlapping complex Gaussian p-body Hamiltonian, $\hbar = 1$, and the normalized maximally mixed trace at $m = N/2$. All norms are annealed Hilbert–Schmidt norms, $\|X\|^2 = \mathbb{E}\,\mathrm{tr}_m\, X^\dagger X$. Let $h = H_m$, $k = H_{m-1}$ have the same couplings and put

$$C_N = \mathbb{E}\,\mathrm{tr}_m\, h^2, \quad B = \mathsf{W}_2 h^2, \quad O = \mathsf{W}_2 A, \quad A = \mathrm{Rem}(k^2) + (2p - m)h^2. \tag{1}$$

Here $\mathrm{Rem}(X) = \sum_i c_i^\dagger X c_i$ and W_2 projects onto ordinary Gaussian second chaos. The quartic core is $\mathcal{E} = B_{22}^f + [h, [h, A]]/4$, with $B_{22}^f = \sum_i (\mathcal{L}^2 c_i)^\dagger \mathcal{L}^2 c_i$ and $\mathcal{L} = [H, \cdot]$ on full Fock space. Set $\lambda_N = p^2/N \to \lambda > 0$, $q_N = e^{-4\lambda_N}$, $C_N \to J_s^2/\lambda$. In formulas below C, λ_N, q_N retain their finite-N values ($C = C_N$). The result is

$$\begin{aligned} \mathsf{W}_2\mathcal{E} &= C[2p(1 - q_N)^2 + \overline{K}_N]B + C\gamma_N O \\ &\quad + 2(q_N - 1)J_h F + 8\lambda_N C q_N^2 \mathcal{S}_N + o_{L^2}(1), \\ \gamma_N &= (3 - 4q_N + q_N^2)/2, \\ \overline{K}_N &= \lambda_N(1 - 10q_N + 9q_N^2) - 16\lambda_N^2 q_N^2. \end{aligned} \tag{2}$$

$\mathcal{S}_N$ inserts the output-rank deficit, and $J_h F = \{h, F\}/2$ is the preceding centered linear-field descendant. Their precise definitions and the assembly are given below. The retained C_N, q_N in the term proportional to p are essential.

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2 Positive product weights and a finite recurrence

At charge M, take independent isotropic Hermitian rank-a, b matrices U, V normalized by $\mathbb{E}U^2 = \mathbb{E}V^2 = I$. Define

$$\pi_{ab}^{(M)}(r) = \mathbb{E}\,\mathrm{tr}_M[P_r(UV)]^\dagger P_r(UV), \qquad \sum_r \pi_{ab}^{(M)}(r) = 1. \tag{3}$$

These are positive, vanish for $r > a + b$, and incorporate all multiplicities. The established covariance/Racah framework [1, 2] gives

$$\sum_r \pi_{ab}^{(M)}(r)\chi_r(n) = \chi_a(n)\chi_b(n), \tag{4}$$

$$\chi_a(n) = {}_4F_3\left[\begin{matrix}-a, a-N-1, -n, n-N-1 \\ -M, M-N, -N-1\end{matrix}; 1\right].$$

This is the previously used composition of covariance maps, not a new probability ansatz. Introduce $x_j = j(N+1-j)$, $P_j(x) = \prod_{\ell=0}^{j-1}(x - x_\ell)$, and

$$c_{nj} = \frac{(-n)_j (n-N-1)_j}{(-M)_j (M-N)_j (-N-1)_j j!}. \tag{5}$$

Since $\chi_r(n) = \sum_{j=0}^n c_{nj} P_j(x_r)$, (4) is triangular. Writing $F_j = \mathbb{E}_\pi P_j(x_r)$ gives the exact recursion

$$F_0 = 1, \qquad F_n = \frac{\chi_a(n)\chi_b(n) - \sum_{j<n} c_{nj} F_j}{c_{nn}}. \tag{6}$$

Raw moments follow by triangular conversion from P_j to x^j. Thus every moment of $X = x_{a+b} - x_r$ through degree six is an explicit rational function of N, M, a, b.

For uniform asymptotics use $\epsilon = N^{-1/2}$, $a = A_0/\epsilon$, $b = B_0/\epsilon$, and $M = 1/(2\epsilon^2) + \delta$, with bounded A_0, B_0, δ. The two charges needed here have $\delta = 0, -1$. Scale the polynomial variable to $Y = \epsilon^3 x_r$. Its roots and input values are

$$\epsilon^3 x_\ell = \ell\epsilon + \ell(1-\ell)\epsilon^3, \tag{7}$$

$$Y_a = A_0 - A_0^2\epsilon + A_0\epsilon^2. \tag{8}$$

In this scaled basis the leading coefficient at degree n is $(-4)^n\epsilon^n + O(\epsilon^{n+2})$. After extracting ϵ^n, all divisors stay away from zero on compact parameter sets. This makes the formal calculation an analytic uniform estimate.

3 Sixth-moment bound and the growing output window

The exact polynomial-ring calculation in `memory_sixth_moment.py` expands (6) through ϵ^{12} over $\mathbb{Q}[A_0, B_0, \delta]$. For each $1 \le j \le 6$, all j negative Laurent coefficients in $\mathbb{E}_\pi(X/N)^j$ vanish. The constant coefficient is $T_j(2A_0B_0)$, where $T_0(t) = 1$, $T_{j+1}(t) = t[T_j(t) + T_j'(t)]$. In particular,

$$T_6(t) = t^6 + 15t^5 + 65t^4 + 90t^3 + 31t^2 + t. \tag{9}$$

The certificate is a polynomial identity, not a parameter sample. The rational analyticity just established makes the remainders uniform, including $A_0 = 0$ or $B_0 = 0$. Consequently

$$\sup_{a,b\le p,\ M=m,m-1} \mathbb{E}_{\pi_{ab}^{(M)}}(X/N)^6 \le K_\lambda. \tag{10}$$

Put $S = a + b - r \ge 0$. Since

$$X = S[N + 1 - 2(a + b) + S] \ge S(N + 1 - 4p), \tag{11}$$

and $N + 1 - 4p > 0$ eventually, both charges satisfy

$$\sup_{a,b\le p} \mathbb{E}_{\pi_{ab}^{(M)}} S^6 \le K_\lambda. \tag{12}$$

Throughout, constants may depend on a compact positive interval of λ_N and on fixed J_s.

Decompose $h = \sum_a h_a$, $k = \sum_a k_a$ using the same rank coordinates. Write $C_a = \mathbb{E}\,\mathrm{tr}_m\, h_a^2$, $\omega_a = C_a/C$, and $n = m + 1$. The preceding transfer calculation gives

$$R_a := \frac{C_a^-}{C_a} = \frac{(m-p)^2 n}{m(mn - x_a)} \le K_\lambda, \qquad \|\,\mathrm{Rem}\,\|_{2,m-1\to 2,m} \le m. \tag{13}$$

For $a < b$, let $Z^+ = P_r\{h_a, h_b\}$ and $Z^- = P_r\,\mathrm{Rem}\{k_a, k_b\}$; for $a = b$ use squares. Set $A_{ab;r} = Z^-_{ab;r} + (2p - m)Z^+_{ab;r}$. Symmetrized independent products have strength at most 4π; Gaussian decoupling gives the same bound for the diagonal input pair, up to an absolute factor. Markov's inequality in (12) gives

$$\sum_{a\le b,\ S>R} \|\mathsf{W}_2 A_{ab;r}\|^2 \le K_\lambda C^2 m^2 R^{-6} \le K_\lambda C^2 p^4 R^{-6}. \tag{14}$$

Here separate upper bounds on the two charges are used only outside a growing window; their large prefactors are explicitly retained.

4 Uniform filter and strong replacement

The exact two-charge contraction gives $\mathsf{W}_2\mathcal{E} = \sum(g_H \mathsf{W}_2 Z^+ + g_A \mathsf{W}_2 A_{ab;r})$. With $d = m - p$,

$$\begin{aligned}
\frac{g_A}{C} &= U_0 + U_a\frac{\kappa_a}{C} + U_b\frac{\kappa_b}{C} + U_r\frac{\kappa_r}{C}, \\
\kappa_j &= w\binom{m-j}{p}\binom{N-m+p-j}{p}, \quad C = \kappa_0, \\
U_0 &= \frac{3d(n+p)}{mn} + \frac{5}{2} - \frac{4d}{m}, \\
U_a &= -\frac{4m+p}{4d} - \frac{(m-n+p)p}{(m-a)(n-a)}, \quad U_b = U_a|_{a\to b}, \\
U_r &= \frac{1}{2} + \left[m + 3n - p - \frac{5(2mn - x_a - x_b)}{4d}\right]\frac{p}{(m-r)(n-r)}.
\end{aligned} \tag{15}$$

For $a = \alpha p, b = \beta p, r = \rho p$, with $0 \le \alpha, \beta \le 1$, $0 \le \rho \le 2$, the uniform expansion is

$$(U_0, U_a, U_b, U_r) = (3/2, -1, -1, 1/2) + \frac{\lambda_N}{p}(8, -5/2, -5/2, 3) + O(p^{-2}). \tag{16}$$

The scaled rational denominators stay nonzero. The exact binomial product gives, uniformly for $0 \le j \le 2p$,

$$|\kappa_{j+1} - \kappa_j|/C \le K_\lambda/p, \quad \kappa_p/C = q_N + O(p^{-1}), \quad \kappa_{2p}/C = q_N^2 + O(p^{-1}). \tag{17}$$

With $u = p - a, v = p - b$, this proves

$$|g_A/C| \le K_\lambda, \qquad |g_A/C - \gamma_N| \le \frac{K_\lambda}{p}(1 + u + v + S). \tag{18}$$

The preceding coherent norm proof supplies the indispensable bound

$$\sum_{a\le b,r} (1 + u + v)^2 \|\mathsf{W}_2 A_{ab;r}\|^2 \le K_\lambda C^2. \tag{19}$$

Different unordered input pairs are orthogonal in Gaussian second chaos; different output ranks are orthogonal in trace norm. Define $E_A = \sum(g_A - C\gamma_N)\mathsf{W}_2 A_{ab;r}$. On $S \le R$, use (18) and (19); on $S > R$, use the uniform bound and (14). Since C remains bounded, this yields

$$\|E_A\|^2 \le K_\lambda[p^{-2} + R^2/p^2 + p^4/R^6]. \tag{20}$$

Choosing $R = p^{3/4}$ proves

$$\boxed{\sum g_A \mathsf{W}_2 A_{ab;r} = C\gamma_N O + O_{L^2}(p^{-1/4}).} \tag{21}$$

This argument does not require tightness in a fixed output window.

5 Assembly and a positive projected core

Define

$$D_N = \sum_a (p-a)h_a, \quad \bar{s}_N = \sum_a (p-a)\omega_a, \tag{22}$$

$$L_{\rm in} = \mathsf{W}_2\{h, D_N - \bar{s}_N h\}, \qquad \mathcal{S}_N = \sum_{a\le b,r} (a+b-r)\mathsf{W}_2 Z^+_{ab;r}. \tag{23}$$

The preceding H-part limit, uniform on compact positive λ_N intervals, reads

$$\begin{aligned} \sum g_H \mathsf{W}_2 Z^+ &= C[2p(1-q_N)^2 + K_{0,N}]B \\ &\quad + 8\lambda_N C(q_N^2 - q_N)\mathsf{W}_2\{h, D_N\} + 8\lambda_N C q_N^2 \mathcal{S}_N + o_{L^2}(1), \\ K_{0,N} &= \lambda_N[1 - 10q_N + 9q_N^2 + 16\lambda_N q_N - 32\lambda_N q_N^2]. \end{aligned} \tag{24}$$

Here $\bar{s}_N - \lambda_N \to 0$ and $\mathsf{W}_2\{h, D_N\} = L_{\rm in} + 2\bar{s}_N B$. The established linear-field bridge is

$$J_h F - 4\lambda_N q_N C L_{\rm in} \longrightarrow 0 \quad \text{in } L^2. \tag{25}$$

Explicitly, $F = \mathsf{W}_1 T_3 - \bar{t}h$, $\bar{t} = \langle h, \mathsf{W}_1 T_3\rangle / C$, where T_3 is the previous cubic derivative singlet. Substitution together with (21) gives (2), since $\overline{K}_N = K_{0,N} + 16\lambda_N^2(q_N^2 - q_N)$. Only the A-part has the rate proved in (21); the assembled remainder is stated as $o_{L^2}(1)$.

Let Π_N be the actual finite-N orthogonal projection onto $\mathrm{span}\{B, O, J_h F\}$. The preceding four-direction Gram is nondegenerate in the limit at every fixed $\lambda > 0$ and gives

$$\|(1-\Pi_N)\mathcal{S}_N\|^2 \longrightarrow V_{\rm new} = \frac{J_s^4(1+q)(1-q^2+8\lambda q)}{\lambda(1+q+4\lambda q)}, \quad q = e^{-4\lambda}. \tag{26}$$

Projecting (2) removes the term of order p exactly. The projection is a contraction, so its action on the remainder vanishes without a rate assumption. Therefore

$$\boxed{\lim \|(1-\Pi_N)\mathsf{W}_2\mathcal{E}\|^2 = \frac{64 J_s^8 q^4(1+q)(1-q^2+8\lambda q)}{\lambda(1+q+4\lambda q)} > 0.} \tag{27}$$

After the double-scaled limit, $\lambda \to 0^+$ gives $1024 J_s^8$. The dimensions are consistent: $A, B, \mathcal{S}_N$ have energy squared, C has energy squared, $J_h F$ and $\mathcal{E}$ have energy to the fourth, and (27) has energy to the eighth.

6 Evidence, scope, and the next calculation

Three saved computation records support the derivation. The symbolic sixth-moment certificate checks 21 vanishing negative powers and all six constant coefficients in an exact polynomial ring. Four complete finite Racah probability tables independently check 28 moments, including order zero. A further 48 parameter sets use exact rational moments followed by 80- and 120-digit evaluations. The filter record checks 130 exact identities against the earlier independent five-word contraction and checks the core assembly algebra.

Finite convergence is not especially fast. For $\lambda = 1/2$, $u = v = 0$, $M = N/2$, the sixth moments of X/N at $p = 16, 32, 64, 128$ are approximately $75.6892, 123.1213, 157.8280, 178.9193$, while their limit is 203. The corresponding exact rational upper bounds on $\mathbb{E}S^6$ have decimal values $166.4086, 180.7793, 190.8031, 196.6133$. These are proxy moments and upper bounds, respectively; the large system calculation does not evaluate the actual S moment. Decimal agreement is not an interval-rounding certificate.

The strategic change is that a growing window suffices for the filter problem. The coarse bound p^4/R^6 is useful at $R = p^{3/4}$, although it cannot establish tightness at fixed R. The coherent cancellation from the preceding norm proof is still used inside the window through (19). We have not proved a full Poisson law, or fixed-window tightness of A, by matching six moments of the ordinary product distribution.

The result completes $\mathsf{W}_2\mathcal{E}$ in the fixed half-filled state. The projected direction is independent of the specified three directions at equal time. The next calculation must restore the low chaos pieces of the fourth-chaos descendants and assemble the raw second memory coefficient Z_2. Extension to a central charge window, energy-resolved joint measures, and finite-time memory remain to be done before a bulk meson interpretation can be extracted.

References

[1] V. K. B. Kota and M. Vyas, *Random Matrix Theory for Transition Strength Densities in Finite Quantum Systems: Results from Embedded Unitary Ensembles*, arXiv:1504.00759, Sec. 2, Eqs. (7)–(10).

[2] R. Koekoek and R. F. Swarttouw, *The Askey-scheme of hypergeometric orthogonal polynomials and its q-analogue*, report 98-17, Sec. 1.2, Eq. (1.2.1).

[3] Alan Day and Chengheng, local project notes (6 November 2026): *The two-charge memory formula*; *Parity-weighted output moments*; *Coherent adjacent-charge cross*; *A positive residual in the quadratic memory sector*. See the accompanying source record for repository paths.